Gambaran Project Akhir

Mahasiswa akan membuat sistem sederhana, tentang “Sistem Manajemen Produk & Order berbasis Microservices”.

Terdiri dari:

- Service 1: User Service
- Service 2: Product Service
- Service 3: Order Service
- API Gateway sederhana
- Komunikasi antar service via HTTP (REST API)

Kebutuhan Sistem

1. Hardware Minimum

   RAM: 4 GB (disarankan 8 GB)

   Storage: 10 GB kosong

   Processor: Dual Core

2. Sistem Operasi

   Windows 10/11 / Linux / MacOS

3. Software Pendukung
   - Python 3.10+
   - pip
   - Postman / Insomnia (testing API)
   - VS Code + Extension Python
   - Git (opsional)
4. Library Python

   Install : *pip install fastapi uvicorn requests*

```
PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> pip install fastapi uvicorn requests
Collecting fastapi
  Downloading fastapi-0.135.2-py3-none-any.whl (117 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 117.4/117.4 kB 1.7 MB/s eta 0:00:00
Collecting uvicorn
  Downloading uvicorn-0.42.0-py3-none-any.whl (68 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 68.8/68.8 kB ? eta 0:00:00
Collecting requests
  Downloading requests-2.33.0-py3-none-any.whl (65 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 65.0/65.0 kB ? eta 0:00:00
Collecting typing-extensions>=4.8.0
  Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Collecting starlette>=0.46.0
  Downloading starlette-1.0.0-py3-none-any.whl (72 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 72.7/72.7 kB ? eta 0:00:00
Collecting annotated-doc>=0.0.2
  Downloading annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Collecting typing-inspection>=0.4.2
  Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Collecting pydantic>=2.9.0
  Downloading pydantic-2.12.5-py3-none-any.whl (463 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 463.6/463.6 kB 4.8 MB/s eta 0:00:00
Collecting click>=7.0
  Downloading click-8.3.1-py3-none-any.whl (108 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108.3/108.3 kB 6.1 MB/s eta 0:00:00
```

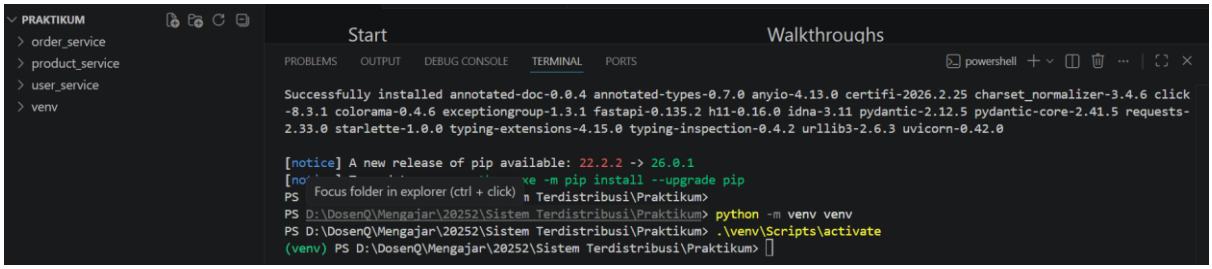
```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71.0/71.0 kB 3.8 MB/s eta 0:00:00
Collecting certifi>=2023.5.7
  Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 153.7/153.7 kB 4.5 MB/s eta 0:00:00
Collecting colorama
  Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Collecting annotated-types>=0.6.0
  Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
Collecting pydantic-core==2.41.5
  Downloading pydantic_core-2.41.5-cp310-cp310-win_amd64.whl (2.0 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.0/2.0 MB 7.5 MB/s eta 0:00:00
Collecting anyio<5,>=3.6.2
  Downloading anyio-4.13.0-py3-none-any.whl (114 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 114.4/114.4 kB 6.5 MB/s eta 0:00:00
Collecting exceptiongroup>=1.0.2
  Downloading exceptiongroup-1.3.1-py3-none-any.whl (16 kB)
Installing collected packages: urllib3, typing-extensions, idna, h11, colorama, charset_normalizer, certifi, annotated-types, annotated
-doc, typing-inspection, requests, pydantic-core, exceptiongroup, click, uvicorn, pydantic, anyio, starlette, fastapi
Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.13.0 certifi-2026.2.25 charset_normalizer-3.4.6 click-8.3.1 co
lorama-0.4.6 exceptiongroup-1.3.1 fastapi-0.135.2 h11-0.16.0 idna-3.11 pydantic-2.12.5 pydantic-core-2.41.5 requests-2.33.0 starlette-1
.0.0 typing-extensions-4.15.0 typing-inspection-0.4.2 urllib3-2.6.3 uvicorn-0.42.0

[notice] A new release of pip available: 22.2.2 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum>
```

Pertemuan 1 – Pengenalan Microservices & Setup

1. Struktur Folder Project

Root Folder **PRAKTIKUM**, didalan ada sub folder dengan nama folder sebagai berikut.

- *user_service*
- *product_service*
- *order_service*



2. Kemudian Setup Virtual Environment

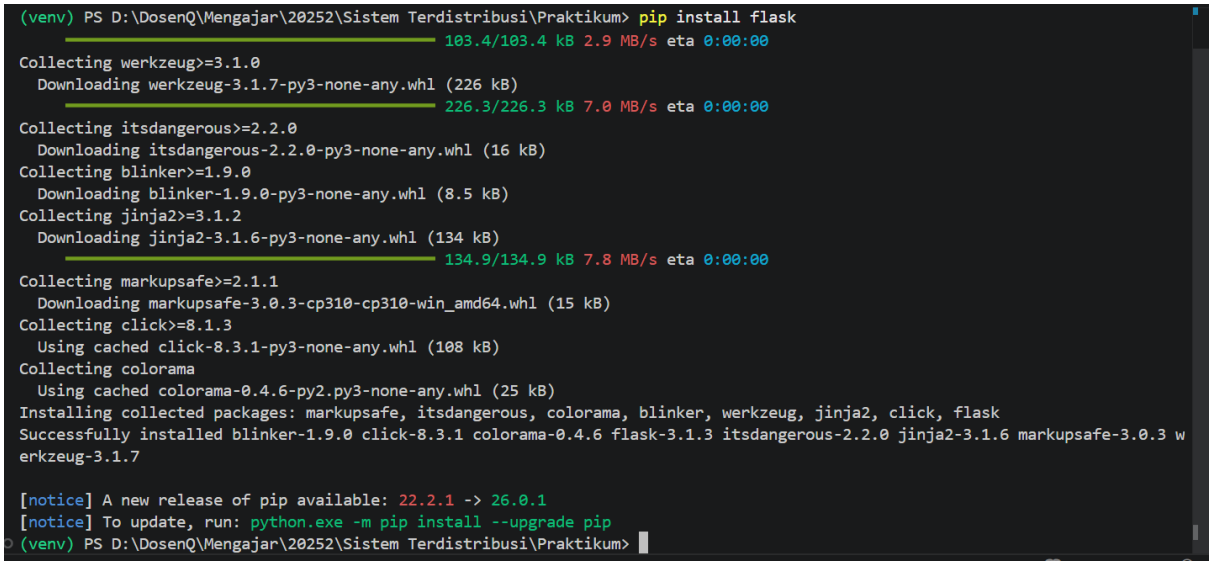
python -m venv venv

3. Aktifkan perintah untuk Windows:

.\venv\Scripts\activate

4. Install Framework Flask

pip install flask



Atau *python -m pip install flask*

```
PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> python -m pip install flask
Collecting flask
  Using cached flask-3.1.3-py3-none-any.whl (103 kB)
Collecting werkzeug>=3.1.0
  Using cached werkzeug-3.1.7-py3-none-any.whl (226 kB)
Collecting jinja2>=3.1.2
  Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Collecting blinker>=1.9.0
  Using cached blinker-1.9.0-py3-none-any.whl (
Requirement already satisfied: click>=8.1.3 in c:\laragon\bin\python\python-3.10\lib\site-packages (from flask) (8.3.1)
Collecting markupsafe>=2.1.1
  Using cached markupsafe-3.0.3-cp310-cp310-win_amd64.whl (15 kB)
Collecting itsdangerous>=2.2.0
  Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Requirement already satisfied: colorama in c:\laragon\bin\python\python-3.10\lib\site-packages (from click>=8.1.3->flask
) (0.4.6)
Installing collected packages: markupsafe, itsdangerous, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.7

[notice] A new release of pip available: 22.2.2 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

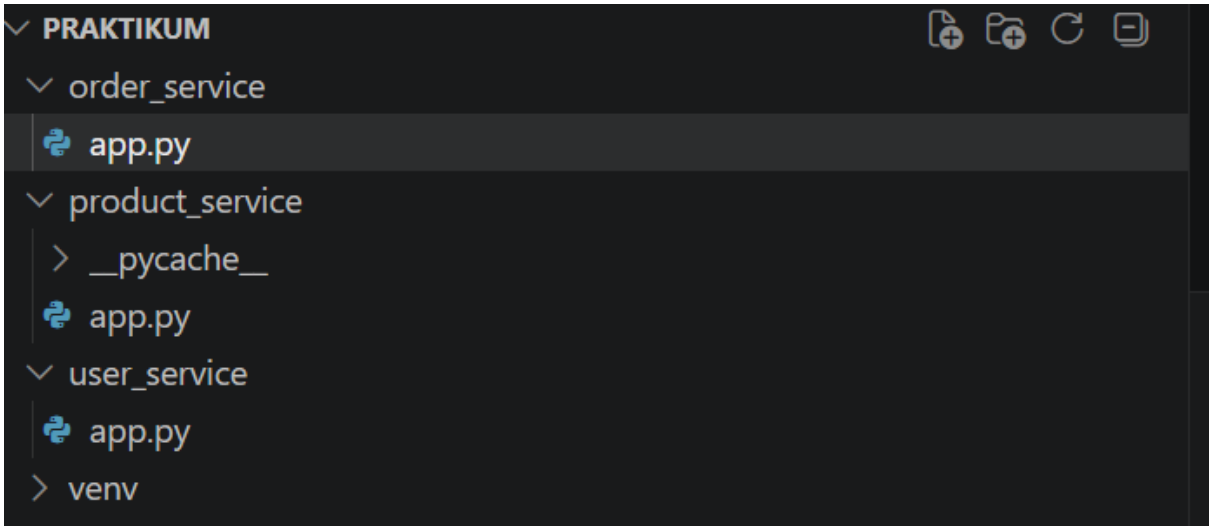
Open folder in new window (ctrl + click)

Cek Flask sudah terinstal apa belum

pip show flask

```
(venv) PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> pip show flask
Name: Flask
Version: 3.1.3
Summary: A simple framework for building complex web applications.
Home-page:
Author:
Author-email:
License:
Location: d:\dosenq\mengajar\20252\sistem terdistribusi\praktikum\venv\lib\site-packages
Requires: blinker, click, itsdangerous, jinja2, markupsafe, werkzeug
Required-by:
(venv) PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum>
```

5. Buat File ***app.py*** dimasing-masing sub folder sehingga akan seperti gambar di bawah ini.



1. **Code untuk order_service/app.py**

   Silahkan ketikan code dibawah ini pada file app.py yang berada di folder order_service.

```
from flask import Flask
```

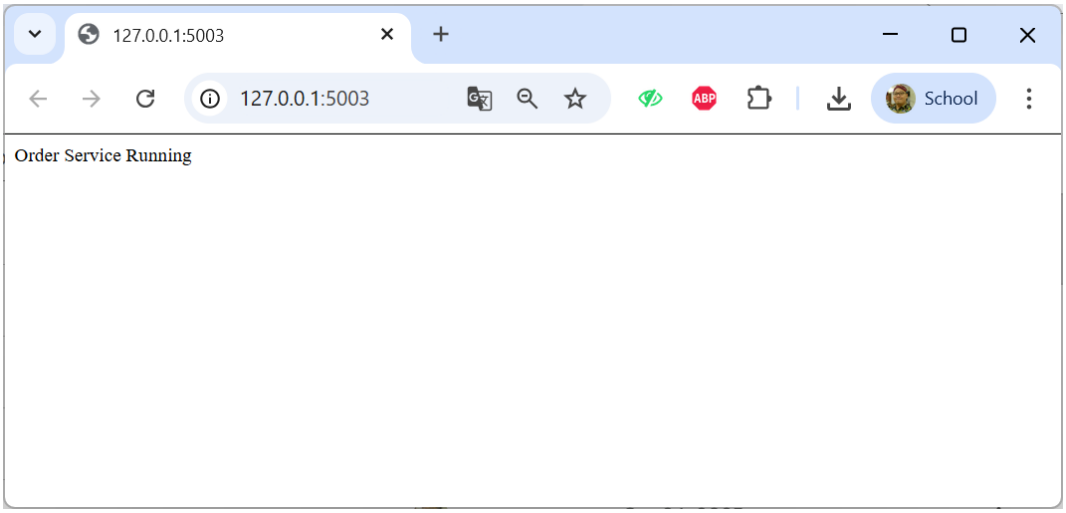
```
app = Flask(__name__)

@app.route('/')
def home():
    return "Order Service Running"

@app.route('/orders')
def orders():
    return {
        "orders": ["Order1", "Order2"]
    }

if __name__ == '__main__':
    app.run(port=5003)
```

Menjalankan order_service/app.py

setalah itu jalankan dengan perintah sebagai berikut, *python order_service/app.py*. Untuk gambarannya bisa dilihat seperti gambar dibawah ini.

```
(venv) PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> python order_service/app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5003
Press CTRL+C to quit
127.0.0.1 - - [30/Mar/2026 09:39:04] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [30/Mar/2026 09:39:04] "GET /favicon.ico HTTP/1.1" 404 -
```

Sedangkan kita cek di browser seperti gambar dibawah ini.



2. **Code untuk product_service/app.py**

   Silahkan ketikan code dibawah ini pada file app.py yang berada di folder product_service.

```
from flask import Flask
```

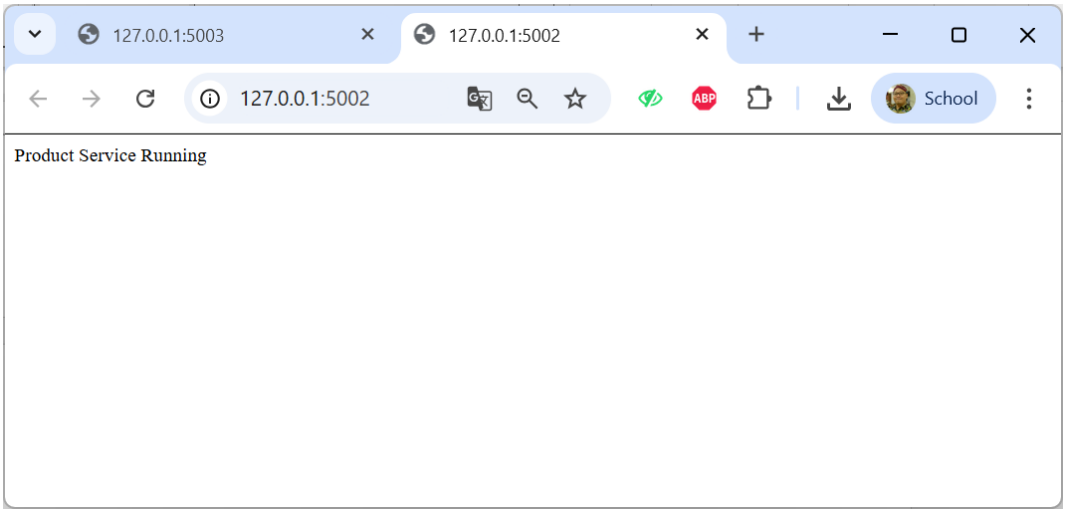
```python
app = Flask(__name__)

@app.route('/')
def home():
    return "Product Service Running"

@app.route('/products')
def products():
    return {
        "products": ["Laptop", "Mouse", "Keyboard"]
    }

if __name__ == '__main__':
    app.run(port=5002)
```

Menjalankan product_service/app.py

Untuk menjalankan perintahnya sebagai berikut *python product_service/app.py*. untuk tampilannya seperti dibawah ini.

```
PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> python product_service\app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5002
Press CTRL+C to quit
127.0.0.1 - - [30/Mar/2026 09:46:15] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [30/Mar/2026 09:46:15] "GET /favicon.ico HTTP/1.1" 404 -
```

sedangkan hasilnya dibrowser seperti gambar dibawah berikut ini.



3. **Code untuk user_service/app.py**

   Silahkan ketikan code dibawah ini pada file app.py yang berada di folder user_service.

```python
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "User Service Running"

@app.route('/users')
def users():
    return {
        "users": ["Admin", "User1", "User2"]
    }

if __name__ == '__main__':
    app.run(port=5001)
```

Menjalankan user_service/app.py

Untuk menjalankan perintahnya sebagai berikut *python user_service/app.py*. Dengan gambaran menjalankan perintahnya seperti dibawah ini.

```
PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> python user_service/app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5001
Press CTRL+C to quit
127.0.0.1 - - [30/Mar/2026 09:48:34] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [30/Mar/2026 09:48:34] "GET /favicon.ico HTTP/1.1" 404 -
```

sedangkan hasilnya dapat kita lihat di browser seperti gambar dibawah ini.